1 Characteristic Functions

1.1 Basic Definitions

Definition 1.1.1

If X is a random variable we define its ***characteristic function (ch.f.)*** by

$$\varphi(t) = Ee^{itX} = E\cos tX + iE\sin tX$$

Theorem 1.1.2

All characteristic functions have the following properties:

(a) $\varphi(0) = 1$,
(b) $\varphi(-t) = \overline{\varphi(t)}$,
(c) $|\varphi(t)| = |Ee^{itX}| \le E|e^{itX}| = 1$
(d) $|\varphi(t+h) - \varphi(t)| \le E|e^{ihX} - 1|$, so $\varphi(t)$ is uniformly continuous on $(-\infty, \infty)$.
(e) $Ee^{it(aX+b)} = e^{itb}\varphi(at)$

1.2 Moments and Derivatives

Theorem 1.2.1

If $E\left|X\right|^n < \infty$, then its characteristic functino is n times differentiable at 0, and $\varphi^n\left(0\right) = E\left(iX\right)^n$. Expanding φ in a Taylor series about 0 leads to

$$\varphi\left(t\right) = \sum_{m=0}^{n} \frac{E\left(itX\right)^m}{m!} + o\left(t^n\right)$$

Theorem 1.2.2

If $E\left|X\right|^2 < \infty$, then

$$\varphi\left(t\right) = 1 + itEX - \frac{1}{2}t^2E\left(X^2\right) + o\left(t^2\right)$$